

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Episiotomy

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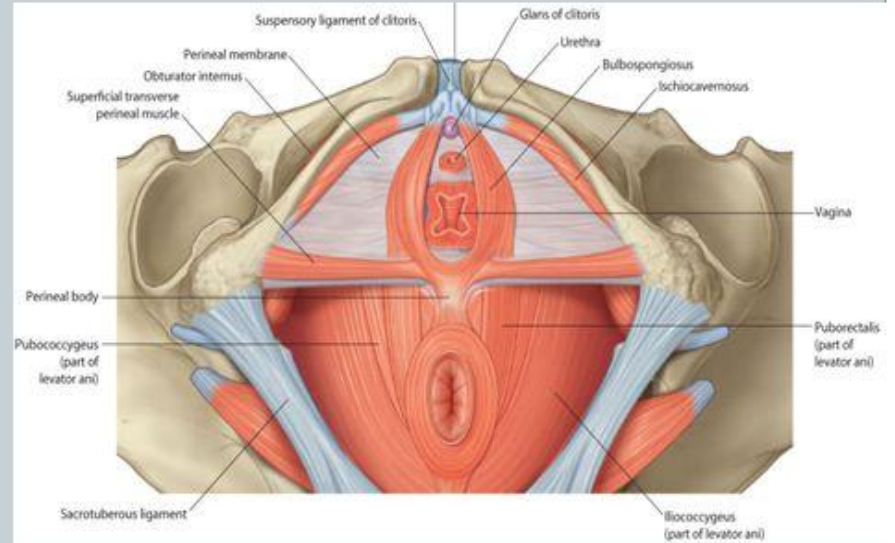
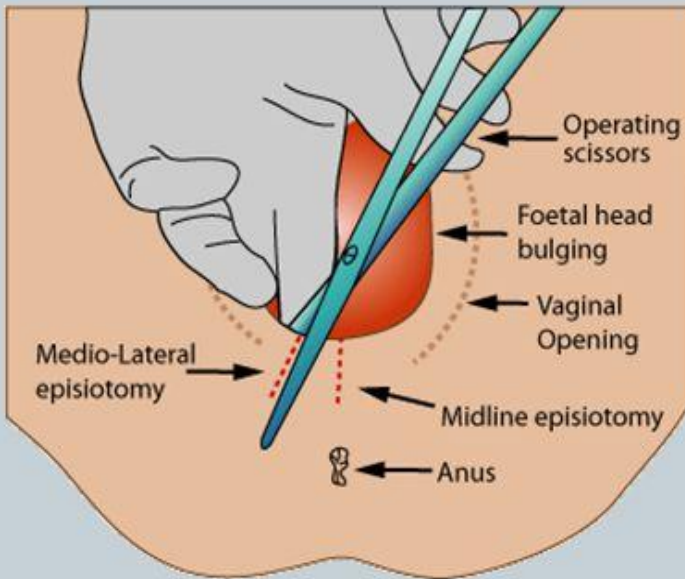
EPISIOTOMY

It is an intentional surgical incision of the perineum and posterior vaginal wall usually done during the second stage of labour to quickly enlarge the vaginal opening for the baby to pass through.

Episiotomy is like a 2nd degree perineal tear and involves cutting the following structures

- **1. The posterior vaginal wall**
- **2.The perineal muscles:**
 - The bulbo-spongiosus and
 - Transverse perineal muscle.
- **3. The perineal skin.**

Episiotomy





Anatomy of Perineum for Perineal repair.mp4

What are the benefits of episiotomy

- **The main aim of doing episiotomy is to avoid spontaneous tearing of the perineum during child birth that may extend and tear the anal sphincter and lead to faecal incontinence.**
- **It has also been suggested that a clean cut incision can be repaired and healed much better than a spontaneous tears.**
- **Unfortunately episiotomy by itself may extend and tear the anal sphincter especially the midline one.**
- **So whether episiotomy reduces, or indeed increases, the chances of anal injury and postpartum anal and sexual dysfunction is difficult to say.**

Indications of episiotomy:

Given the lack of proven benefit to episiotomy, routine use of the procedure should be abandoned; and targeted or restricted use of episiotomy is appropriate.

So the deliver provider should perform an episiotomy whenever perineal tear is felt eminent.

Indications:

- Rigid perineum.
- Shoulder dystocia.
- Fetal distress.
- An instrumental or breech delivery.
- Women who has previous pelvic floor surgery

Types of episiotomy:

There are four main types of episiotomy incisions:

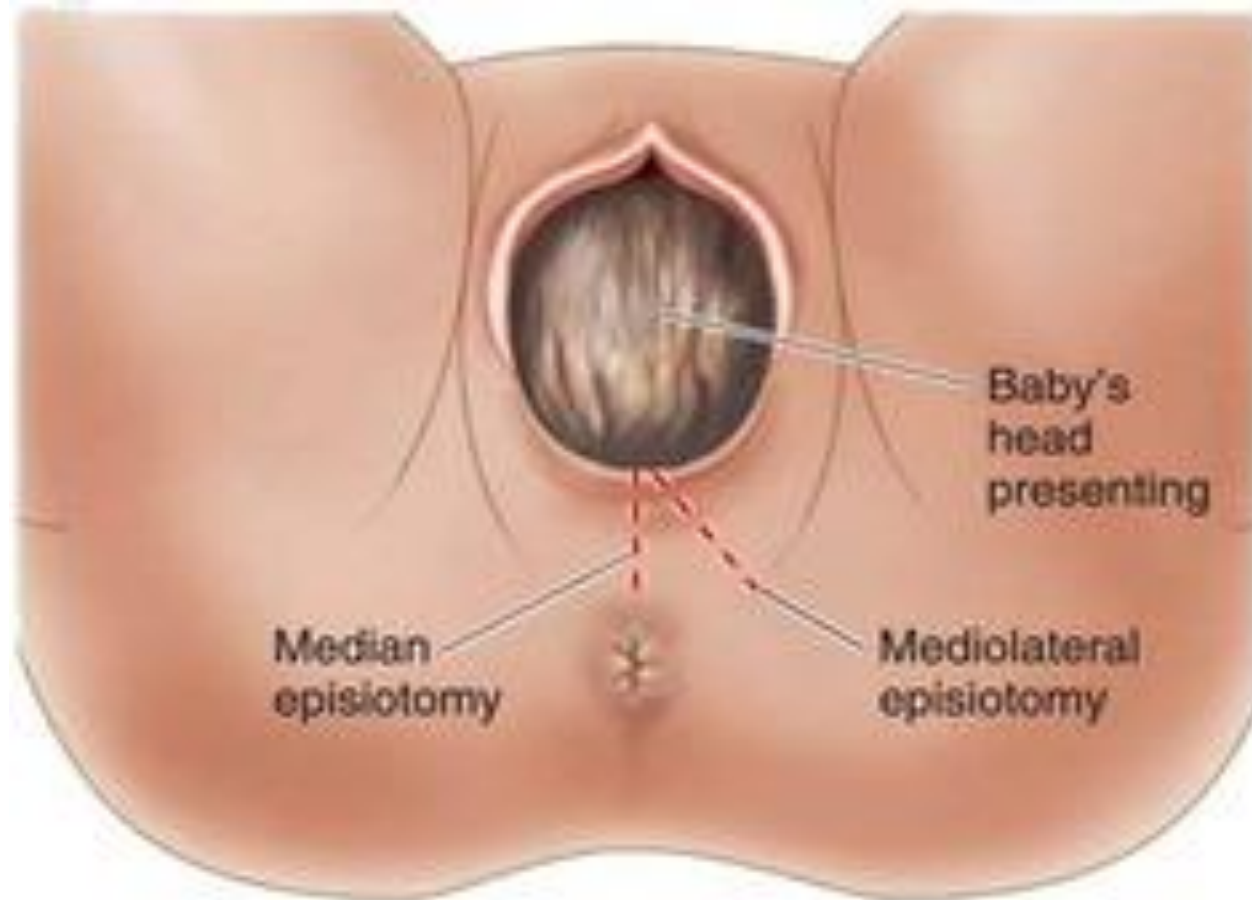
1. Midline (median):

- A midline episiotomy starting from the center of the fourchette and extends on the posterior side along the midline for 2.5 cm towards the anus (mainly used in USA).

2. Mediolateral:

- A mediolateral episiotomy starting from the mid-point of the fourchette going laterally toward the buttock about 2.5cm away from the anus (about 45° angle away from midline).
- This type is preferable and used in most parts of the world.

Midline and mediolateral episiotomy:



3.J-shaped:

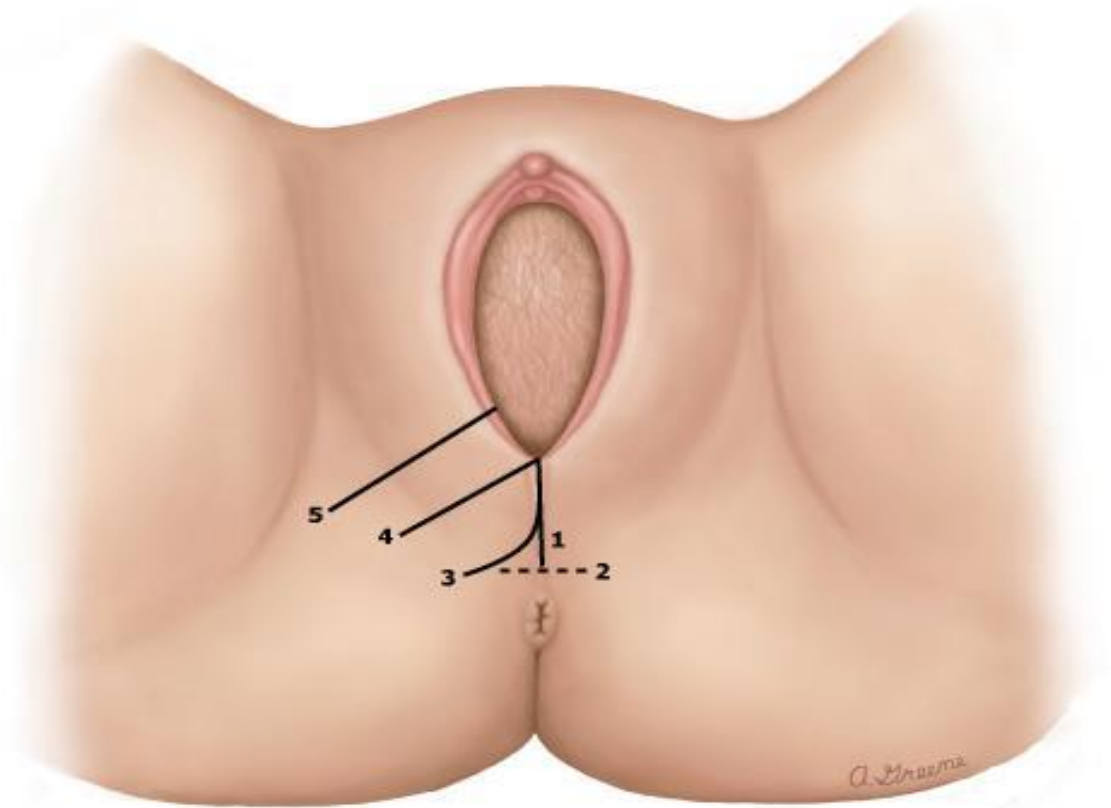
- The incision begins in the centre of the fourchette and is directed posteriorly along the midline for about 1.5 cm and then directed downwards and outwards along the 5 or 7 o'clock position to avoid the internal and external anal sphincter.
- This procedure is also not widely practiced.

4.Lateral:

- The incision starts from about 1 cm away from the centre of the fourchette and extends laterally.
- Drawbacks include the chance of injury to the Bartholin's duct, therefore it is strongly discouraged.

Types of episiotomy

1. Medline
2. modified midline.
- 3.J-shaped.
4. Mediolateral.
5. Lateral



How to perform an episiotomy

Anesthesia:

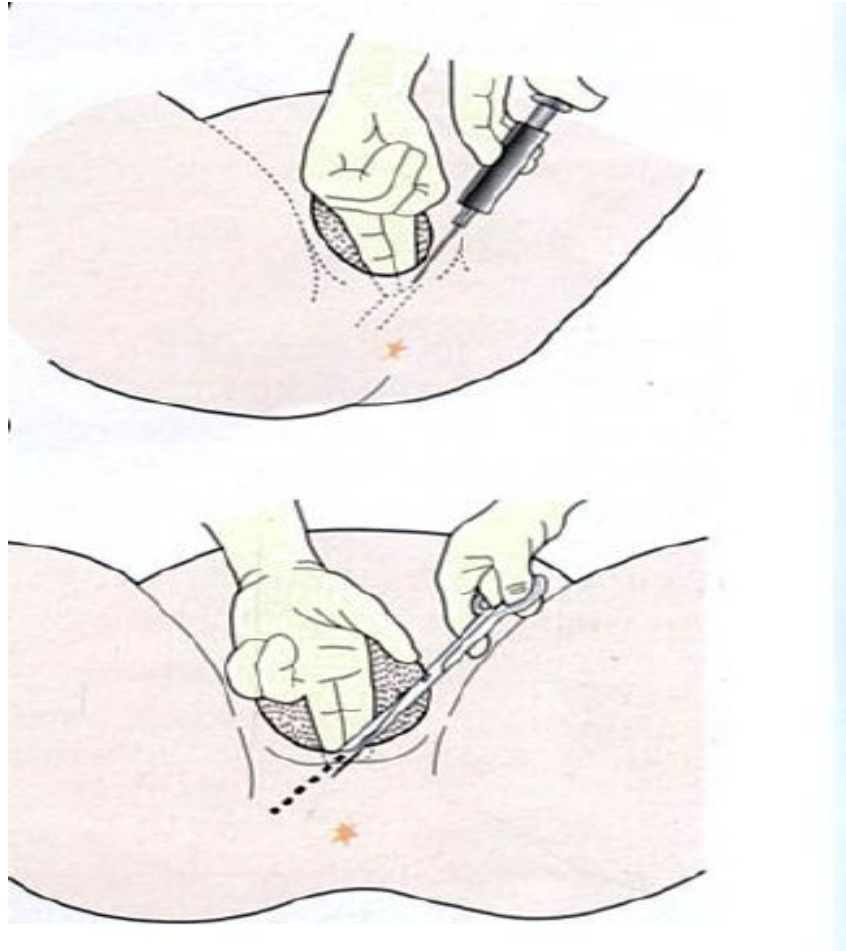
-in cases of normal vaginal delivery it can be done under local infiltration of the perineum.

-In cases of instrumental deliveries it should be done under pudendal block or epidural anesthesia.

Technique:

- A sharp scissors is used to make a single incision about 3–4 cm depending on the size of the perineum.
- The cut should be done with **crowning of the fetal head**.
- Two fingers are placed between the fetal head the vagina to protect the fetal head.
- The episiotomy must be made **in a single cut**.
- If it is enlarged by several small cuts , a zigzag incision will be produced which will be difficult to repair.
- For mediolateral episiotomy **the incision should begin in the midpoint at the fourchette and cut laterally at an angle of 60 degree away from the anus(when the perineum is over stretched by the fetal head).**

Performing episiotomy:





Performing an Episiotomy.mp4

Midline versus mediolateral Episiotomy

The advantages of a midline episiotomy include :

- Easier to repair with less blood loss.
- Heals well and quicker compared with the mediolateral episiotomy.
- There is less pain in the postpartum period.
- The incidence of dyspareunia is less.
- The main disadvantage of a midline episiotomy is the increased risk for tears that extend into or through the anal muscles.
- (so it is not widely used).

Mediolateral Episiotomy

- The primary advantage of a mediolateral episiotomy is that the risk for anal muscles injury is much lower so it is the preferable type in most parts of the world.

Repair of episiotomy

- Repair of any perineal laceration, including episiotomy, serves the following 2 important functions:

1.Homeostasis.

2.Tissue re-approximation for lacerated vaginal mucosa, soft tissue and muscles.

-Accordingly, delivering providers must have a comprehensive understanding of perineal anatomy.

Suture materials:

- Two types of suture material has been used for repair of episiotomy:
 1. Chromic catgut (Natural), (now out of use).
 2. Synthetic absorbable sutures like: polyglycolic acid (Dexon) and polyglactin 910 (vicryl)
- The use of absorbable synthetic sutures for repair of perineal trauma is associated with:
 - Less perineal pain and analgesic use.
 - Less dehiscence and re-suturing.
 - But increased suture removal, when compared with catgut.

Now there are rapid-absorption polyglactin which is the most appropriate suture material for perineal repair.

In most instances 2-0 is enough for repair of soft tissue of the perineum.

Principles of repair

- **Suture as soon as possible following delivery to reduce bleeding and risk of infection.**
- **Check equipment and count swabs prior to commencing the procedure and count again following completion of the repair.**
- **Good lighting is essential to visualize and identify the structures involved.**
- **Ask for more experienced assistance if in doubt regarding the extent of trauma or structures involved.**

- Difficult trauma should be repaired by an experienced operator in theatre under regional or general anaesthesia.
- Insert an indwelling catheter for 24 hours to prevent urinary retention.
- Ensure good anatomical alignment of the wound and give consideration to cosmetic results.
- Rectal examination after completing the repair will ensure that suture material has not been accidentally inserted through the rectal mucosa.
- Following completion of the repair, inform the woman regarding:
 - 1.The extent of trauma.
 - 2.Discuss pain relief.
 - 3.Diet (drink a lot of water and high fiber diet to avoid constipation).
 4. Hygiene (proper cleaning the area especially after urination and defecation)
 - 5.The importance of pelvic-floor exercises.

Methods of repair:

- **Perineal trauma is conventionally repaired in three layers**
- **A thorough examination of the perineum should always be performed to evaluate for extension to a third- or fourth-degree laceration.**
- **First insert a vaginal tampon or swab to prevent blood from the uterus from obscuring the field.**
- **The repair is initiated by placing a suture approximately 5-10mm proximal to the apex of the laceration within the vagina to secure any retracted blood vessel.**
- **The suture is secured with a knot, and the repair is continued in a continuous fashion to the level of the hymenal ring.**
- **The vaginal mucosa and sub-mucosal tissue must be sufficiently re-approximated with the continuous suture, and all dead space must be closed.**

- **Careful attention must be directed to preventing sutures from passing deep enough to traverse the rectal mucosa.**
- **Next, the hymen is re-approximated with the suture in a non-locked fashion.**

2.-Then the muscular tissue of the perineal body is re-approximated by interrupted stitches.

3.Finally, the skin is closed using continuous subcutaneous or interrupted trans-cutaneous techniques.



Episiotomy épisiotomie technique et matériels nécessaire.mp4

Medications:

- The goals of pharmacotherapy are to prevent complications and infections and reduce morbidity.
- Infection increases the risk of perineal repair breakdown, particularly (third- or fourth-degree) lacerations which may result in incontinence of stool or flatus, sexual dysfunction, or recto-vaginal fistula,
- **The use of prophylactic antibiotics in this setting is highly indicated.**

Complications of episiotomy:

- If the episiotomy is performed too far laterally it will not increase the diameter of the vulval outlet but may cause damage to the right Bartholin's gland which cause a decrease in vaginal lubrication or cyst formation.**
- Extension to third and fourth degree tear.**
- Bleeding which can be heavy.**
- Infection.**
- Wound breakdown**
- Dyspareunia.**

Thank you

